Lecture 3:
Stocks and bonds

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MN3101 Corporate Finance and Control

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Lecture Overview

Bond Valuation
- Bond Terminology
- Bond Pricing
- Bond Price and Yield to Maturity
- Time and Interest Rate Effects
- The Yield Curve
- Corporate Bonds and Risk of Default

Stocks Valuation
- Stocks and Stock Trading
- Valuing Common Stocks
- The Dividend Discount Model
- Repurchases and the Total Payout Model
- Valuation Based on Comparable Firms



Bond Valuation

Topics Covered

- Bond Terminology
- Bond Pricing
- Bond Price and its Yield-To-Maturity
- Bond Price Behavior (Time and Interest Rate Effects)
- The Yield Curve
- Corporate Bonds and Risk of Default



Reading

- HRWJJ - Chapter 5
- BMAE - Chapter 3



Bond Terminology

A bond is a security sold to raise money from investors today in exchange for the promised future payment.

Face value (Par value or Principal value)	Final payment at maturity. Also used to compute interest payments.
Coupon rate	Annual interest rate, as a percentage of face value.
Coupon payment	The interest payments made to the bondholder.
Maturity date	The final repayment date.

Coupon Rate

The **coupon rate is NOT the discount rate** used in the present value calculations.

By convention, the coupon rate is expressed as an APR, which merely tells us what cash flow the bond will produce. The coupon rate is set by the issuer, as stated on the bond certificate.



Coupon Payment (CPN)

$$CPN = \frac{Face\ Value \times Coupon\ Rate}{N.of\ Coupon\ Payments\ per\ Year}$$

A $1000 bond with a 10% coupon rate and semiannual payments will pay coupon payments every six months of:

$$CPN = \frac{1000 \times 10\%}{2} = \$50$$

Bond Maturity

Gov't bonds in 2016

- Spain: 50-year bonds (€3 billion, 3.45%)
- Ireland: 100-year
- Belgium: 100-year

Corporate bonds

- Walt Disney 'Sleeping Beauty' in 1993 (7.55%, semi-annually)
- Canadian Pacific Railway in 1889
- Safra Republic Holdings S.A in 1997 (callable)

Source:
http://www.wsj.com/articles/long-dated-bonds-find-favor-as-investors-try-to-dodge-negative-yields-1462978982
http://www.wsj.com/articles/SB876150541352487500
http://www.theglobeandmail.com/globe-investor/why-cps-old-time-bondholders-have-a-big-say-in-the-future/article4105862/
http://www.cpr.ca/en/community/holiday-train

Zero-Coupon Bonds

A zero-coupon bond is a bond that does not make coupon payments. The only payment the bondholder receives is the face value of the bond on the maturity date.

Prior to its maturity date, the price of a zero-coupon bond is always less than its face value: pure discount bonds.



Coupon Bonds

A coupon bond makes regular coupon interest payments. If the coupon rate is fixed, coupon payments are equal throughout the life of the bond: level coupon bonds.

Callable bonds	Convertible bonds	Inflation-indexed	Floating rate bonds



Until recently interest rates were often quoted as some spread over LIBOR. The London Interbank Offered Rate (LIBOR) is the rate at which banks can borrow from each other in the London interbank market. A panel of banks submitted their rates for different currencies and maturities.

In the 2010–2012 **LIBOR scandal**, bankers at major financial institutions (including Deutsche Bank, Barclays, Citigroup, JPMorgan Chase, and the Royal Bank of Scotland) had colluded to manipulate LIBOR. Changes to report based on existing transactions and FCA in charge. Decline in reliability due to decline in active underlying markets to support LIBOR. End of LIBOR in 2024.

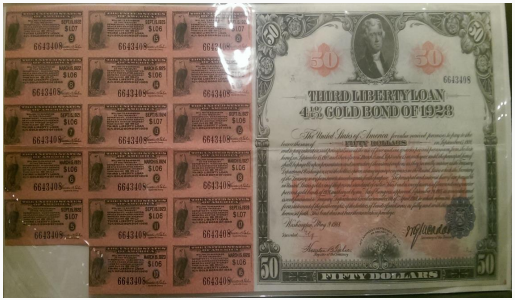
Today Risk-free-Referecence Rates (RFRs) are used.

SONIA	SOFR
for GBP	for USD

SONIA (Sterling Overnight Index Average) is compiled by the Bank of England. It is based on actual transactions and reflects the average of the interest rates that banks pay to borrow sterling overnight from other financial institutions and other institutional investors.

Bond Certificate 1918 by the US Treasury

A Coupon Bond



- Issued: 1918
- Due: 1928
- Par: $50
- Rate: 4.25%
- Semi-annual
- Used to fund World War I

Bond Trading

Bond trading takes place over-the-counter (OTC), a dealer market. The number of bond issues far exceeds the number of stock issues (companies can have a dozen different bonds and one stock), bond trades are much larger in size, and trades are less frequent than stock.

Lack of transparency: since 2002 dealers have to report trades and sale prices through TRACE (Trade Reporting and Compliance Engine). Still if there are few trades or no trades prices are not transparent.

Bond Pricing

The price of a bond is the present value of all cash flows generated by the bond (i.e., coupons and face value) discounted at the required rate of return.

$$P_B = \frac{CPN_1}{(1+r)^1} + \frac{CPN_2}{(1+r)^2} + \cdots + \frac{CPN_N}{(1+r)^N} + \frac{PAR}{(1+r)^N}$$

$$P_B = \frac{CPN}{r} \times \left[1 - \frac{1}{(1+r)^N}\right] + \frac{PAR}{(1+r)^N} \quad (2)$$

Bond Pricing: Example 1

What is the price of a 7.25% annual coupon bond, with a $1,000 face value, which matures in 3 years? Assume a required return of 0.35% and the time now is May 2013.

$$P_B = \frac{72.5}{0.0035} \times \left[1 - \frac{1}{(1+0.0035)^3}\right] + \frac{1000}{(1.0035)^3}$$

$$P_B = \$1205.56$$

Example 1: Semi-annual Coupons

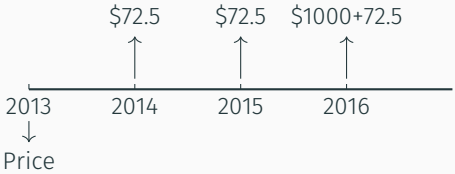
What is the price of the bond if the coupons are paid semi-annually?

$$P_B = \frac{36.25}{0.00175} \times \left[1 - \frac{1}{(1 + 0.00175)^6}\right] + \frac{1000}{(1.00175)^6}$$

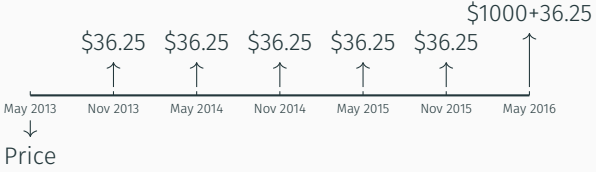
$$P_B = \$1205.74$$

Timeline of Bond Cash Flows

Annual coupon payments



Semi-annual coupon payments



Yield to Maturity

The yield to maturity (YTM) of a bond is the discount rate that sets the present value of the promised bond payments equal to the current market price of the bond.

YTM = y = r = IRR = required rate of return. YTM moves in line with the market interest rate.

Calculating YTM of a Bond: Example 2

A five-year, $1000 bond with 5% coupon rate and semi-annual coupons is currently trading for a price of $957.35. What is the bond's YTM?

Solution

$$\$957.35 = \frac{\$25}{YTM} \times \left[1 - \frac{1}{(1+YTM)^{10}}\right] + \frac{\$1000}{(1+YTM)^{10}}$$

By trial-and-error:

- When YTM = Coupon rate = 2.5%: Price = $1000
- Try YTM = 3.5%: Price = $916.83
- Interpolation: $\frac{(YTM-0.025)}{(0.035-0.025)} = \frac{(1000-957.35)}{(1000-916.83)}$
- YTM $\approx$ 3.01%. The price function is convex, so interpolating between two trial rates overstates the yield slightly; solving exactly gives 3%, a 6% APR (3% $\times$ 2)

By Excel: = rate(nper , pmt , pv , fv) = rate(10, 25, -957.35, 1000) = 3%

A Bond's Price and Its YTM

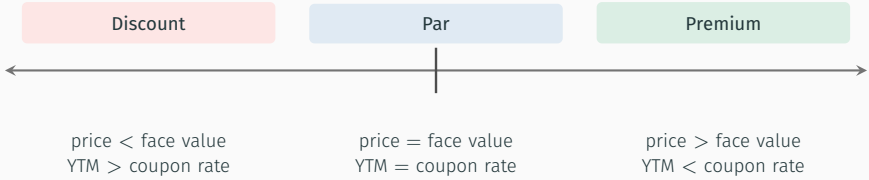
As bond yields rise, bond prices will fall, and vice versa.

Because we can convert any price into a yield, and vice versa, prices and yields are used interchangeably. Indeed, bond traders generally quote bond yields rather than prices.



Dynamic Behaviour of Bond Prices

Coupon bonds may trade at prices lower, higher or the same as their face values because of the effects of time and interest rates.



Zero-coupon bonds always trade at a discount: investors receive a face value which is higher than the price paid for the bond.

Time and Bond Prices: Zero-coupon Bonds

Suppose you purchase a 30-year, zero-coupon bond with a YTM of 5% and a face value of $100. What is the price of this bond five years later if the bond's YTM remains at 5%?

The bond will initially trade for

$23.14

Five years later

$29.53

The bond price is higher not because of the lower yield but less time to maturity

Time and Bond Prices: Coupon Bonds

Bond prices rise as the next coupon payment gets closer and then drop after it has been paid.

This result also holds for coupon bonds. The pattern of price changes over time is slightly more complicated because as time passes most of the cash flows get closer but some disappear as the coupons get paid.



The Effect of Time on Bond Prices: Example 3

Consider a 30-year bond with a 10% coupon rate (annual payments) and a face value of $100.

1. What is the initial price of this bond if it has a 5% YTM?
2. If the YTM is unchanged, what will the price be immediately BEFORE and AFTER the first coupon is paid?



Example 3: Solution

$$P_{Initial} = \frac{10}{0.05} \times \left[1 - \frac{1}{1.05^{30}}\right] + \frac{100}{1.05^{30}} = \$176.86$$

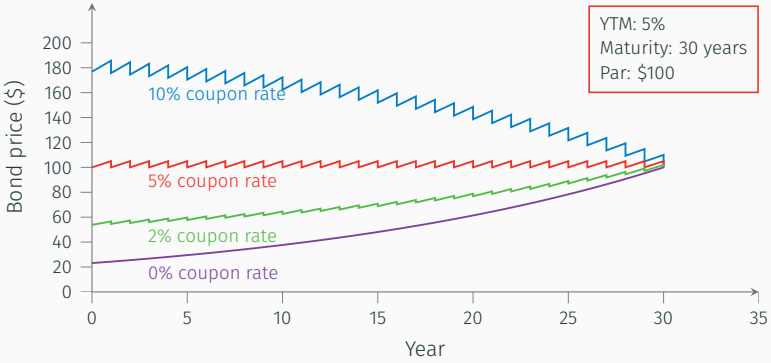
$$P_{Before} = 10 + \frac{10}{0.05} \times \left[1 - \frac{1}{1.05^{29}}\right] + \frac{100}{1.05^{29}} = \$185.71$$

$$P_{After} = \frac{10}{0.05} \times \left[1 - \frac{1}{1.05^{29}}\right] + \frac{100}{1.05^{29}} = \$175.71$$

Example 3: Some Discussions

Immediately BEFORE the coupon payment	The bond price is higher than the initial price: no changes in the number of coupon payments but an investor does not have to wait as long to receive the first coupon payment.
Immediately AFTER the coupon payment	The bond price drops by the amount of the coupon ($10): the new bond owner will not receive this first coupon payment.
After the first payment	The price becomes lower than the initial price: fewer coupon payments remaining, investors pay less for the bond.

Time and Bond Prices: Graphical Presentation

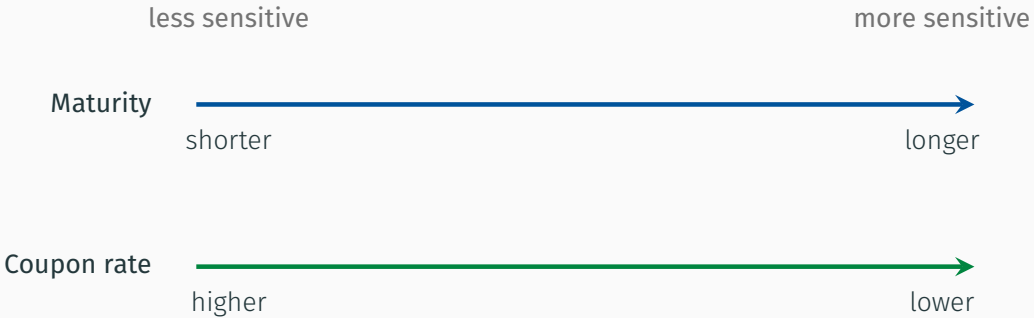


Interest Rate Changes and Bond Prices

When interest rates rise, investors demand high yields (higher discount rates), reducing the price of bonds.

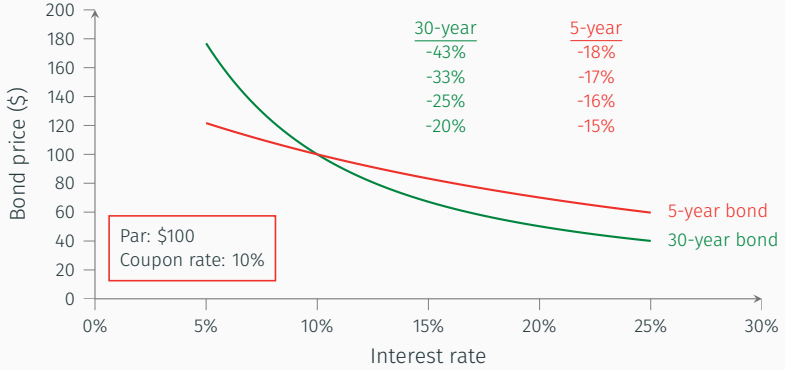
As interest rates in the economy fluctuate, the yield that investors demand from bonds will also change, as do the bond prices.

The Interest Rate Sensitivity of Bond Prices



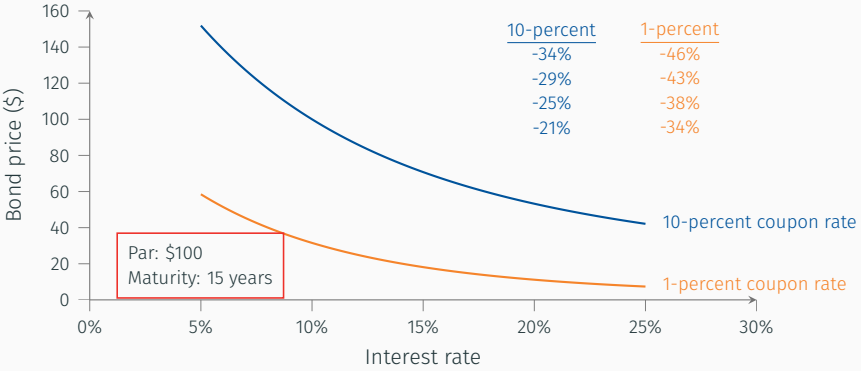
Sensitivity of Bond Prices to Interest Rate

Different Maturities



Sensitivity of Bond Prices to Interest Rate

Different Coupon Rates



Sensitivity of Bond Prices: Time and Interest Rate

In actuality, bond prices are subject to both time effect and interest rate changes.

Time effect	Bond prices converge to the bond's face value.
Interest rate effect	Bond prices move up and down due to changes in bond yields.

Time and Interest Rate: Example 4

Consider a 15-year zero-coupon bond and a 30-year coupon bond with 10% annual coupons ($100 face value). By what percentage will the price of each bond change if its YTM increases from 5% to 6%?

Example 4: Solution

15-year zero-coupon bond

$$P_{15,5\%} = \frac{100}{1.05^{15}} = \$48.10$$

$$P_{15,6\%} = \frac{100}{1.06^{15}} = \$41.73$$

$$\Delta P_{15} = \frac{(41.73 - 48.10)}{48.10} = -13.3\%$$

30-year coupon bond

$$P_{30,5\%} = \frac{10}{0.05} \times \left[1 - \frac{1}{1.05^{30}}\right] + \frac{100}{1.05^{30}} = \$176.86$$

$$P_{30,6\%} = \frac{10}{0.06} \times \left[1 - \frac{1}{1.06^{30}}\right] + \frac{100}{1.06^{30}} = \$155.06$$

$$\Delta P_{30} = \frac{(155.06 - 176.86)}{176.86} = -12.3\%$$

The Yield Curve: Description

A plot of relationship between bond YTM and time to maturity is known as the yield curve.

A yield curve can also be described as the term structure of interest rates.



The Yield Curve: Graph



Source: Bank of England nominal government liability curve, 17 September 2026 https://www.bankofengland.co.uk/statistics/yield-curves

Corporate Bonds

For corporate bonds, the issuer may default; this risk is known as credit risk.

Government bonds tend to be safe in terms of default (at least those denominated in local currency). Investors pay less for bonds with credit risk than they would for default-free (additional yield = default premium).



Example 5

Andrew Corporation issues a one-year zero coupon bond with a face value of $1000. Potential bondholders expect there to be a 50% chance that the bond will repay its face value in full (if the economy becomes stronger) and a 50% chance that the bond will default, and bondholders will receive only 90% of the face value at the end of one year (if the economy becomes weaker). As a result, investors buy the bond for $920. Assume that the one-year risk-free interest rate is 2%.

1. What is the expected and promised YTM of the bond?
2. What is the default risk premium required by investors for this bond?



Example 5: Solution

$$\text{Expected cash inflow} = 0.5 \times 1000 + 0.5 \times 900 = \$950$$

$$\text{Expected YTM} = (950/920) - 1 = 3.26\%$$

$$\text{Promised YTM} = (1000/920) - 1 = 8.70\%$$

$$\text{Risk premium} = 3.26\% - 2\% = 1.26\%$$

The 8.70% promised YTM is the most investors will receive (no default). If the company defaults, investors will receive only \$900, a return of $(900/920) - 1 = -2.17\%$. The average return (the cost of debt) is $3.26\% = 0.5 \times 8.70\% + 0.5 \times (-2.17\%)$.

Bond Ratings

The safety of most corporate bonds can be judged from bond ratings provided by Moody's, Standard and Poor's, Fitch Ratings or other bond rating firms.

Highest quality	Aaa (Moody's), AAA (S&P, Fitch)
Investment-grade bonds	Baa, BBB and above
Speculative / junk / high-yield bonds	Below Baa

Protection against default risk: seniority, security and protective covenants

- Extra debt restriction
- No senior new debt
- Gearing limit

Stocks Valuation

Topics Covered

- Stocks and Stock Trading
- Valuing Common Stocks
- The Dividend Discount Model
- Repurchases and the Total Payout Model
- Valuation Based on Comparable Firms

Reading

- HRWJJ - Chapter 5
- BMAE - Chapter 4
- Pinto, J.E., Robinson, T.R. and Stowe, J.D. (2019) Equity Valuation: A Survey of Professional Practice. *Review of Financial Economics* 37, 219-233. https://doi.org/10.1002/rfe.1040
- Brav, A., Graham, J.R., Harvey, C.R. and Michaely, R. (2005) Payout Policy in the 21st Century. *Journal of Financial Economics* 77, 483-527. https://doi.org/10.1016/j.jfineco.2004.07.004

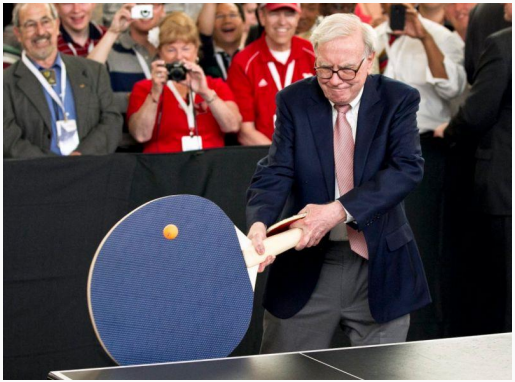


Common Stock

Common stock	A certificate of capital ownership in a publicly held corporation.
Dividend	Periodic cash distribution from the firm to the shareholders.
Capital gain	Earnings come from selling stocks.

Classes of Common Stocks

A primary reasons for firms creating multiple classes of stock is to do with control of the firm. Firms can raise equity capital by issuing nonvoting or limited-voting stock.



- Class B (BRK.B) shares have 1/10,000th of the voting rights of Class A (BRK.A) shares
- B shares have value 1/1,500th of Class A shares

Ford, General Motors, Alphabet (Google)

Ford	Class B represents less than 10% of the number of shares outstanding but has 40% voting power.
GM	Class E is to partly finance acquisition of Electronic Data Systems (1984) and class H for Hughes Aircraft (1985).
Alphabet (Google)	Class C (GOOG) has no voting rights; class A (GOOGL) has one vote each; class B has ten votes each (which do not trade in the public market).

Over-the-counter	Market participants look up the prices quoted and trade with dealers (dealer market).
Electronic trading system	Market participants post orders over computer networks and trading is done automatically without human intervention when orders are matched ('crossed').

Brokers, Dealers and Designated Market Makers (Specialists). Highly regulated.

Stock Price Listings

Honeywell International Inc. (HON) NYSE

206.46 +0.40 (+0.19%)

Prev close	206.06	Day's range	204.26–207.48
Open	206.49	52wk range	195.87–260.28
Volume	7,042,200	Market cap	65.4bn
Avg vol (3m)	3,813,487	P/E (ttm)	7.94
Beta	0.901	EPS (ttm)	26.01
		Div & yield	2.80 (1.36%)

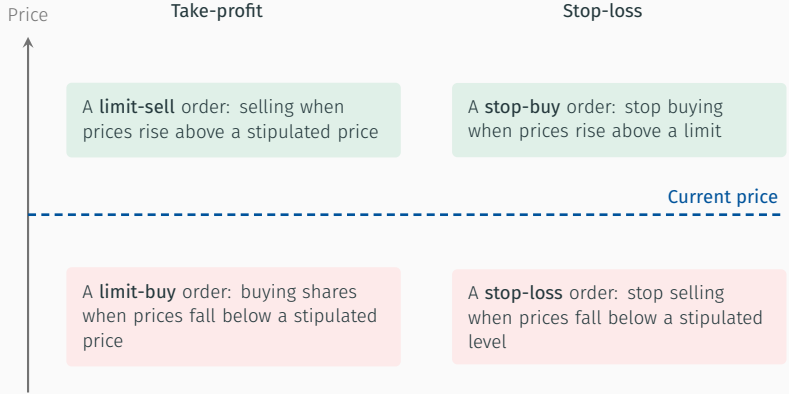
Source: https://uk.finance.yahoo.com [19 September 2026]

Types of Orders

Limit orders	An order to buy or sell shares at a predetermined price, to be executed when the market price reaches the requested price.
Market orders	An order to buy or sell shares at the best currently available market price.



Types of Limit Orders



Common Stock Valuation

The value of a stock is the present value of its current and expected future cash flows.

A common stock is more difficult to value in practice than a bond:

- Promised cash flows are unknown in advance
- Common stocks have no maturity
- The rate of return that the market requires is unobservable

Stock Prices: One-year Investment Horizon

Two potential sources of cash flows from owning a stock:

- Dividend
- Capital gain

Assuming for simplicity that an investor buys a stock today and sells it at the end of the year and all dividends are also paid at the end of the year.

$$P_{0(1)} = \frac{D_1 + P_1}{(1 + r_e)} \quad (1)$$

Eq.(1) can be transformed into (2):

$$r_e = \frac{D_1}{P_0} + \frac{P_1 - P_0}{P_0} \quad (2)$$

- The first term on the right-hand side is the dividend yield and the second term is the capital gain rate
- The sum is the total return of the stock
- If the stock offered a higher return than other securities with the same risk, investors would sell other investments and buy the stock, driving up the current price of the stock

A 2-year Investor

Suppose an investor planned to hold the stock for two years, then this investor would receive dividends in both year 1 and year 2 before selling the stock. Setting the stock price equal to the present value of the future cash flows in this case implies:

$$P_{0(2)} = \frac{D_1}{(1+r_e)} + \frac{(D_2+P_2)}{(1+r_e)^2} \quad (3)$$

We can continue this process for any number of years by replacing the final price with the value that the next holder of the stock would be willing to pay. Doing so leads to the general dividend-discount model:

$$P_{0,N} = \frac{D_1}{(1+r_e)} + \frac{D_2}{(1+r_e)^2} + \cdots + \frac{D_N + P_N}{(1+r_e)^N} \quad (4)$$

If it is possible to hold the stock forever, we can let N go to infinity:

$$P_{0,N} = \frac{D_1}{(1+r_e)} + \frac{D_2}{(1+r_e)^2} + \cdots = \sum_{n=1}^{\infty} \frac{D_n}{(1+r_e)^n} \quad (5)$$

Case 1: Zero Growth

For a zero growth common stock, the dividend is constant through time:

$$D_1 = D_2 = D_3 = \text{constant}$$

So the value of the stock is:

$$P_0 = \frac{D}{(1+r)} + \frac{D}{(1+r)^2} + \frac{D}{(1+r)^3} + \cdots$$

The stock can be viewed as an ordinary perpetuity, and the per-share value is thus given by:

$$\boxed{P_0 = \frac{D}{r} \quad (6)}$$

Case 2: Constant Growth DDM

Another simplification of the DDM is to assume that dividends grow at a constant rate (Gordon Growth Model):

$$D_1 = D_0(1+g)$$
$$D_2 = D_1(1+g) = D_0(1+g)^2$$
$$D_3 = D_2(1+g) = D_0(1+g)^3$$

$$P_0 = \frac{D_1}{(1+r)} + \frac{D_1(1+g)}{(1+r)^2} + \frac{D_1(1+g)^2}{(1+r)^3} + \cdots$$

$$\boxed{P_0 = \frac{D_1}{(r-g)} = \frac{D_0(1+g)}{(r-g)} \quad (7)}$$

Example 1

Suppose Edison Company plans to pay £2.30 per share in dividends in the coming year. If the required rate of return is 7% and dividends are expected to grow by 2% per year in the future, how much should Edison's stock be selling for?

Solution

$$
\begin{aligned}
P_0 &= \frac{D_1}{(r-g)} \\
&= \frac{2.30}{(0.07-0.02)} \\
&= £46.00
\end{aligned}
$$

Example 2

Suppose Blue Skies Company just paid a dividend of £0.50 per share. It is expected to increase its dividend by 2% per year. If the market requires a return of 15% on assets of a similar risk level taken by the company, how much should the stock be selling for?

Solution

$$\begin{aligned} P_0 &= \frac{D_0(1+g)}{(r-g)} \\ &= \frac{0.50(1+0.02)}{(0.15-0.02)} \\ &= £3.92 \end{aligned}$$

Example 3

Suppose Gordon Company will pay a dividend of £4 per share in the coming year. Investors require a 16% return on companies such as Gordon. Gordon's dividend is expected to increase by 6% every year. What is the value of Gordon's stock today? What is the value in four years?

Solution

$$P_0 = \frac{D_1}{(r-g)} = \frac{4}{(0.16-0.06)} = £40.00$$

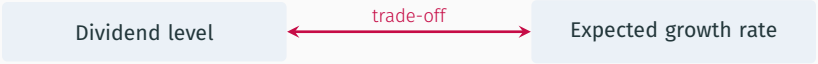
$$P_4 = \frac{D_5}{(r-g)} = \frac{D_1(1+g)^4}{(r-g)} = \frac{4(1+0.06)^4}{(0.16-0.06)} = £50.50$$

$$P_4 = P_0(1+g)^4 = 40.00(1+0.06)^4 = £50.50$$

Stock price will grow at the same constant rate as dividend.

Dividends vs. Investment and Growth

To maximise its share price, a firm would like to increase both the dividend level and expected growth rate as shown in Eq.(7).



However, there is often a trade-off: growth requires investment, which reduces dividend payout. The constant growth model will help to gain insight into this trade-off.

Dividend Payout

If the dividend payout rate (DPR) is defined as the fraction of a firm's earnings that the firms pay as dividends each year, the firm's dividend per share at date t is as follows:

$$D_t = \frac{\text{Earnings}_t}{\text{Shares outstanding}_t} \times \text{DPR}_t = \text{EPS}_t \times \text{DPR}_t \quad (8)$$

The firm can increase dividend in 3 ways:

1. By increasing its earnings
2. By increasing its dividend payout rate
3. By reducing its shares outstanding

To explore the trade-off between growth in earnings (1) and dividend (2), let's suppose that the number of shares outstanding (3) is fixed.

$$\Delta\text{Earnings}_{(t,t+1)} = \text{Invest.}_{(t+1)} \times \text{Return on Invest.}_{(t+1)} \quad (9)$$

$$\text{Invest.}_{(t+1)} = \text{Retained Earnings}_t$$

$$\text{Retained Earnings}_t = \text{Earnings}_t \times \text{Retention Rate}_t \quad (10)$$

$$\text{Earnings Growth}_{(t,t+1)} = g = \frac{\Delta\text{Earnings}_{(t,t+1)}}{\text{Earnings}_t}$$

$$g = \text{Retention Rate}_t \times \text{Return on Invest.}_{(t+1)} \quad (11)$$

Eq. (11) shows that a firm can increase its growth rate by retaining more earnings. However, the firm will have to reduce its dividend, as shown in Eq. (8).

If a firm wants to increase its share price, should it...

cut dividend and invest more?

or

cut investment and increase its dividend?

Example 4

Sports Wear expects to have earnings per share of €7 in the coming year. Rather than reinvest these earnings and grow, the firm plans to pay out all of its earnings as a dividend. With these expectations of no growth (assuming that Sports Wear does not buy back shares and raises funds for investment by issuing more shares or increasing debt), the firm's current stock price is €55.

Suppose Sports Wear could cut its dividend payout rate to 70% for the foreseeable future and use the retained earnings to open new stores. The return on its investment in these new stores is expected to be 15%. Assuming the required rate of return from investors is unchanged, what would the current stock price be?

Example 4: Solution

First, estimate the required rate of return (the firm's cost of capital):

$$r_e = \frac{D_1}{P_0} + g = \frac{7}{55} + 0\% = 12.7\%$$

If the firm reduces its DPR to 70%, then:

$$\begin{aligned} D_1 &= 7 \times 70\% = 4.9 \\ g &= (1 - 70\%) \times 15\% = 4.5\% \\ P_0 &= \frac{D_1}{r_e - g} = \frac{4.9}{12.7\% - 4.5\%} = €59.6 > €55 \end{aligned}$$

Cutting the firm's dividend in favour of growth raised the firm's stock price. But is this always the case?

Example 5: Unprofitable Growth

Suppose Sports Wear decides to cut its dividend payout rate to 70% to invest in new stores as in Example 4 but now the return on this new investment is 10% rather than 15%. Given its expected earnings per share of €7 and its cost of capital of 12.7%, what will happen to the firm's current share price in this case?

Solution

$$g = (1 - 70\%) \times 10\% = 3\%$$

$$P_0 = \frac{D_1}{r_e - g} = \frac{4.9}{12.7\% - 3\%} = €50.4 < €55$$

The new investment has negative NPV: return on new investment < return required by investors, so the share price will fall if the firm cuts its dividend to make this new investment

Profitable vs. Unprofitable Growth

Cutting the firm's dividend to increase investment will raise the stock price if, and only if, the return on new investment exceeds the return required by investors.

The effect of cutting dividend to grow crucially depends on the return on new investment. If new investment is expected to earn only the return required by investors, reinvesting earnings does not add value to stock price (even though it will lead to earnings growth).



Growth and Business Life Cycle

Successful young firms often have very high earnings growth rates; these firms tend to retain all earnings for investment.

The constant growth DDM cannot be used.

As they mature, these firms grow slower, earnings exceed their investment needs and they begin to pay dividends.

Case 3: Differential Growth

However, we can use the general form of the DDM to value these firms. Assume that dividends will grow at a certain rate in the foreseeable future (g_1) and then grow at a different rate (g_2) thereafter.

To value a differential growth stock:

1. Estimate the future dividends in the foreseeable future and calculate the present value of these dividends (from year 1 to N)
2. Estimate the future stock price (P_N) when the stock becomes a constant growth stock
3. Calculate the total present value of (1) and (2)

Case 3: Differential Growth Timeline



$$D_1 = D_0(1+g_1)$$

$$D_2 = D_1(1+g_1) = D_0(1+g_1)^2$$

$$P_N = \frac{D_{N+1}}{r-g_2}$$

$$D_N = D_{N-1}(1+g_1) = D_0(1+g_1)^N$$

$$D_{N+1} = D_N(1+g_2) = D_0(1+g_1)^N(1+g_2)$$

$$P_{0,g_1} = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \cdots + \frac{D_N}{(1+r)^N} = \frac{D_1}{(r-g_1)}\left[1 - \frac{(1+g_1)^N}{(1+r)^N}\right]$$

$$P_{0,g_2} = \frac{P_N}{(1+r)^N} = \frac{D_{N+1}}{(r-g_2)(1+r)^N}$$

$$\boxed{P_0 = \frac{D_1}{(r-g_1)}\left[1 - \frac{(1+g_1)^N}{(1+r)^N}\right] + \frac{1}{(1+r)^N}\frac{D_{N+1}}{(r-g_2)} \quad (12)}$$

P_N: a terminal value in the DDM

Differential Growth: Example 6

A common stock has just paid a dividend of £3. The dividend is expected to grow at 7% for 3 years, then it will grow at 5% forever. The required rate of return is 11%. What is the current stock price?

Solution

$$
\begin{aligned}
P_0 &= \frac{D_1}{(r-g_1)}\left[1-\frac{(1+g_1)^N}{(1+r)^N}\right]+\frac{1}{(1+r)^N}\frac{D_{N+1}}{(r-g_2)} \\
&= \frac{3\times 1.07}{(0.11-0.07)}\left[1-\frac{(1+0.07)^3}{(1+0.11)^3}\right]+\frac{1}{(1+0.11)^3}\frac{3\times 1.07^3\times 1.05}{(0.11-0.05)} \\
&= 8.37+47.03 = \text{£}55.4
\end{aligned}
$$

Limitations of the Dividend Discount Model

The DDM values the stock based on a forecast of the future dividends paid to shareholders. There is a tremendous amount of uncertainty, and small changes in the assumed dividend growth rate can lead to large changes in the estimated stock price.

Forecasting dividends require forecast earnings, dividend payout rate and future shares outstanding:

- Earnings depend on interest expenses, which depend on firm debt
- Shares outstanding and payout rate depend on whether firm uses some earnings to repurchase shares
- Because **borrowing and share repurchase decisions are at management's discretion**, they can become more difficult to forecast

Share Repurchase

In a share repurchase, a firm uses excess cash to buy its own shares.

More cash to buy back shares, less cash to pay dividends. Buying back shares reduces shares outstanding, which increases earnings and dividends on a per-share basis.

In the DDM, a stock is valued from a perspective of a single shareholder, discounting the dividends the shareholder is expected to receive.



Total Payout Model

The total payout model values all of the firm's equity rather than a single share.

An alternative method that might be more reliable when a firm repurchases shares is the total payout model. Total payouts = dividends + share repurchases: discount the total payouts and then divide the present value of the total payouts by the current number of shares outstanding.

We can apply different scenarios of growth rates as the DDM – the only change is that we use total dividends, and share repurchases and use the growth rate of total earnings rather than earnings per share.

Total Payout Model: Example 7

Alpha Industries has 108.5 million shares outstanding and expects earnings at the end of this year of $430 million. Alpha plans to pay out 55% of its earnings in total (paying 30% as a dividend and using 25% to buy back shares). If Alpha's earnings are expected to grow at 7% per year and the payout rate remains constant, what is Alpha's share price? Assuming the firm's cost of equity is 12%.

Solution

$$\text{Total payouts} = 0.55 \times \$430 \text{ million} = \$236.5 \text{ million}$$

$$\text{PV of Future Total Payouts} = \$236.5 \text{ million} \div (0.12 - 0.07) = \$4.73 \text{ billion}$$

$$\text{Price} = \$4.73 \text{ billion} \div 108.5 \text{ million} = \$43.59$$

Valuation Based on Comparable Firms

Price-to-earnings ratio	P/E
Market-to-book ratio	M/B
Price-to-sales ratio	...

But firms are not identical. The usefulness of this valuation approach depends on the nature of the differences between firms (growth rates, risk and payout rates).

Example 8

Suppose a furniture manufacturer, Miller Corporation, has EPS of \$1.5. If the average P/E of comparable furniture stocks is 21, estimate the stock value for Miller using the P/E as a valuation multiple. What are the assumptions underlying this estimate?

Solution

$P_0 = \$1.5 \times 21 = \$31.5$

This estimate assumes that Miller will have similar future risk, payout rates and growth rates to comparable firms in the furniture industry.

Thank You!

Thank you for listening.

For issues or questions come to my office hours.

